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Fatigue: Tough days at work change your prefrontal metabolites

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New measurements of the metabolite and neurotransmitter glutamate in prefrontal cortex after a day of hard work indicate that it may be a brain marker of mental fatigue, re-energising searches for the biological roots of fatigue.

It's early evening. You have just got home after a full day at work focused on a really complex problem. You had planned to

make yourself a slow-cooked ragù for dinner. Perfect. Yet, right now, the idea of exerting all that effort and waiting so many

hours for it to be ready seems much less appealing. You know you can get a good pizza effortlessly delivered to your door in

15 minutes, so why not do that instead? This benign every-day example belies a major challenge in psychological and neuroscientific research^{1,2}. Why is it that when we become mentally fatigued, we avoid hard tasks and seek immediate gratification? In this issue of *Current Biology*, Wiehler *et al.*³ offer a tantalising new answer: doing something mentally demanding for a long period of time leads to a build-up of the metabolite glutamate in the brain, making it harder for us to be patient, or put in effort, for reward.

Biological theories have long proposed that there may be physical changes in the brain or body that occur when we do mentally demanding behaviours for extended periods of time — what we experience as mental fatigue. Much of this work focused on the idea that mental exertion might use up energy reserves, such as glucose⁴. This ‘depletion’ would then make you more averse to effort or less patient in waiting for rewards, because of an impaired ability of the brain to function without sufficient energy. However, the weight of evidence has not supported either the role of glucose or the depletion of other resources as directly impacting levels of mental fatigue^{1,5}.

The failure to support such accounts pushed focus away from understanding the biological root causes of fatigue. Researchers instead proposed psychological accounts^{3,6}. These theories centred on the idea that ‘cognitive control’ — the constant monitoring of stimuli and actions in tasks which cannot be automated⁷ — is perceived as costly and effortful. Repeatedly exerting control makes an ongoing behaviour seem less valuable and increases the desire to switch to something else, with fatigue being the by-product of this process. But whilst it is clear that such ‘opportunity-cost’^{6,8,9} processes influence behaviour, they don’t answer questions at the core of the effects of fatigue. Why does doing one demanding task become more costly over time, and why does it transfer onto decisions about other completely different behaviours or rewards^{1,10}? In essence, psychological accounts struggle to explain why the fatigue caused by working all day makes you less willing to cook the ragù.

Wiehler *et al.*³ propose a new way of reconciling psychological and biological accounts. They suggest that fatigue might arise, not because of depletion, but because taxing a brain area consistently leads to an accumulation of metabolites within it. Specifically, mentally demanding tasks that require cognitive control could lead to an accumulation of glutamate in lateral prefrontal cortex (IPFC), a region known for its role in cognitive control processes¹¹. This accumulation would make IPFC function less efficiently, which would impact other cognitive control processes, such as economic decisions about whether to exert effort or wait for rewards.

To test this, two groups of people undertook an experiment lasting 6.5 hours, almost the same as a day at work. One group performed hard versions of two different types of cognitive control task designed to induce fatigue, which either taxed working memory or the ability to discriminate stimuli based on learned rules. The second group performed easier versions of otherwise identical tasks. This means the groups were closely matched, but only the hard versions should induce significant fatigue. Interspersed with performing these cognitive control tasks, both groups performed economic decision tasks that involve cognitive control. These tasks included making decisions about willingness to exert physical effort (whether to ride a bike at different intensities), cognitive effort (whether to perform harder or easier versions of the cognitive control tasks) and patience (how long they were willing to wait to receive a larger reward). In addition, all participants underwent three magnetic resonance spectroscopy (MRS) scans — a type of magnetic resonance imaging (MRI) scan that detects brain metabolites — while doing blocks of these tasks, at the beginning, middle, and end of the day. With these experiments, Wiehler *et al.*³ were able to compare changes in glutamate levels, and their diffusion, in IPFC and a control region in visual cortex, between groups across the three scans.

As expected, the group that performed the hard version of the cognitive control tasks became less patient, and more averse to exerting both cognitive and physical effort, across the experiment, compared to the

group that had the easy versions. Strikingly, the group that performed the harder cognitive control tasks all day had higher glutamate concentrations and higher glutamate/glutamine diffusion in IPFC at the end of the experiment compared to the group with the easy versions, with no differences in a control region (Figure 1). Thus, continued exertion of high levels of cognitive control appears to result in higher levels of glutamate specifically in IPFC, lower willingness to exert effort into similar cognitive control processes or physically demanding tasks, and less patience to wait for more reward.

These findings are even more compelling when placed in the context of functional (f)MRI studies that measure localised changes in blood oxygenation in the brain during tasks. Such studies have shown that the IPFC response is linked to overtraining in athletes¹² and reduced patience after a day of work¹³, and builds during physically effortful tasks¹⁴. This is all consistent with the idea that performing demanding tasks that require IPFC leads to a build-up of glutamate, in turn making it harder to keep performing tasks that will further tax the IPFC.

This evidence of a biological substrate of mental fatigue is important, because work-induced fatigue impacts not only trivial decisions, but also performance and motivation during crucial work activities¹⁵. Jobs with high risks for performance failures, such as the sustained cognitive control required of pilots or surgeons, may need very different mitigations if there is biological, versus purely psychological, causes. These findings may also have significance for clinical conditions where fatigue is a major feature^{16,17}, including chronic fatigue syndrome and functional disorders, for which there is often no clear or obvious physical manifestation. The work of Wiehler *et al.*³ highlights that there may be many more candidate neural substrates that need to be examined within such conditions before concluding there is no biological cause.

These results are an exciting first step towards a rich, fully fledged metabolic account of mental fatigue, but there are a number of hurdles to be overcome. First, there are limitations to even the most sophisticated MRS protocols. Not all metabolites can be measured

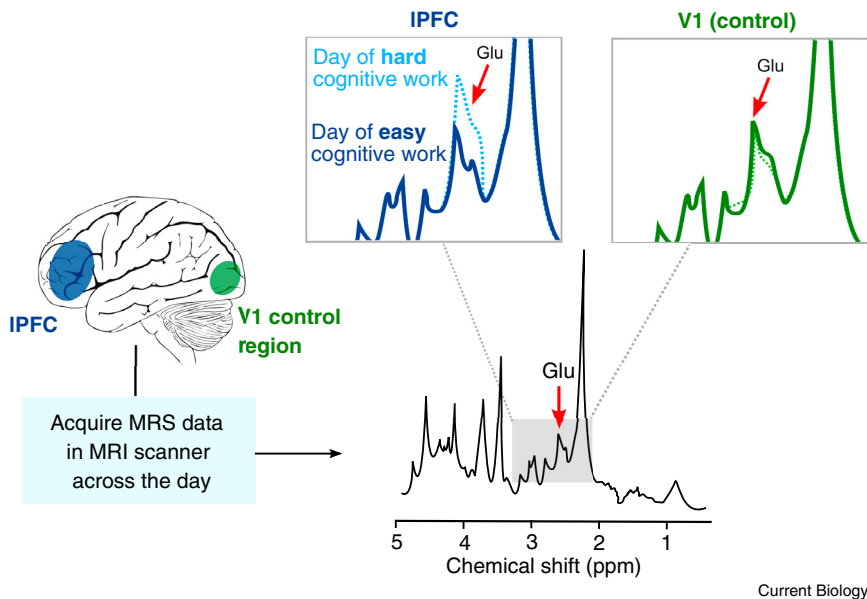


Figure 1. The brain metabolite glutamate increases in IPFC after a day of hard cognitive work.

Measuring brain metabolites using magnetic resonance spectroscopy reveals that continued exertion of cognitive control for high demanding tasks results in higher levels of glutamate (Glu) in lateral prefrontal cortex (IPFC) at the end of the day, compared to low demanding tasks. No changes in glutamate were observed in a visual cortex control region (V1).

simultaneously, and it is possible that other metabolites, such as GABA, may also be accumulated or depleted alongside glutamate when people become mentally fatigued. Second, MRS can only measure metabolites from small parts of the brain at one point in time. It is thus unclear whether other brain areas important for cognitive control and decision-making^{18,19} might show similar or different patterns of changes in glutamate or other metabolite concentrations. Third, important questions need to be resolved regarding the mechanisms by which our willingness to exert cognitive control depends on how much glutamate there is in IPFC — how does the brain know this? Thus, in the future both technological and conceptual advances are needed to offer a fully characterised neurometabolic account of how daylong work leads to reduced patience and effortful exertion.

Finally, fatigue is a complex and multi-dimensional construct, with different brain regions and circuits involved across different tasks and clinical conditions^{16,17}. Recent evidence shows that both sensations of fatigue and the effects of fatigue on motivation can change both on

a very short-term momentary basis, as well over longer timescales as in this study^{12,20}. Future work will need to establish whether all these different aspects of fatigue relate to the same biological substrate in IPFC, or whether this is just one of many phenomena that underlie exhaustion and its impact on decision-making.

The work of Wiehler *et al.*³ paves the way for a resurgence in investigating the biological roots of mental fatigue and offers a reason why we may too often switch plans to getting a takeaway.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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Floodplain ecology: A novel wetland community of the Amazon

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An expedition to the upper estuarine reaches of the Amazon River reveals intriguing overlap of tropical mangrove wetlands with riverine floodplain forests. This newly discovered type of forested wetland assemblage may provide a uniquely process-rich carbon hotspot.

Finding patterns among the earth's impressive collection of habitats is a fundamental goal of ecosystem ecology and biogeography, and it often takes just a single discovery to train ecologists to look for more. In a new study published in this issue of *Current Biology*, Angelo Bernardino *et al.*¹ reveal a novel type of forested wetland assemblage in which mangroves overlap with freshwater forested wetlands. The discovery of this mixed wetland community along the upper estuarine reaches of the lower Amazon floodplain in Brazil adds a new flavor to estuarine science and floodplain ecology alike, and raises questions about its potential for increased societal value, especially in regard to carbon storage.

Mangrove wetlands are known to occupy at least 140,260 km² of intertidal land area globally², where they provide habitat for wildlife and fish, protection from waves and wind, and nutrient uptake and storage from bi-directional water flows where oceans meet upland river and overland drainage. Likewise, tidal freshwater forested wetlands span an unknown area globally, but occupy river edges where tidal range is broad or funneled³. Recent advancement of a 'blue carbon' concept^{4,5}, which focuses on carbon cycling and storage among tidal wetlands not previously considered in terrestrial or oceanic

carbon accounting, has raised awareness of mangrove and tidal freshwater forested wetland value as carbon neutrality frameworks are adjudicated. What if we were to find a habitat where both types of tidal forested wetlands exist? Would such overlap provide synergistic value to society? This new study by Angelo Bernardino *et al.*¹ describes the distribution, structural complexity, and basic biogeochemical attributes of just such a community — a mangrove wetland mixed with obligate freshwater trees along the upper estuarine reaches of the lower Amazon floodplain in Brazil. Given that both systems possess biogeochemical efficiencies that lead to high carbon storage, the mixed mangrove–floodplain forest community described by Bernardino *et al.*¹ may provide the makeup for a new wetland blue carbon hotspot.

Discovery, science advancement, and landscape extent of mangrove wetlands has always revolved around the concept of astronomical tide⁶, distinguishing mangroves from pulsed hydroperiod dynamics of more inland floodplain forests. In the Amazon Basin, flood pulsing from the Río Amazonas and the Río Negro creates vast floodplain forests known as *várzea* (flooded by nutrient-rich, light-colored water) and *igapó* (flooded by

nutrient-poor or rainfall-sourced water), respectively. However, past studies in Amazonia's floodplain forests have understandably overlooked the relatively small area associated with the upper estuary.

In Dr. Wolfgang Junk's seminal floodplain text, *The Central Amazon Floodplain: Ecology of a Pulsing System*⁷, mangroves are defined simply as those "...characterized by a predictable polynomial flood pulse resulting from tidal influence" (p. 6), with only limited extrapolation of overlapping estuarine tidal forest types. As it turns out, Bernardino *et al.*¹ discovered that mixed mangrove–floodplain forests offer larger areas of hydrologic hybridization than previously thought. Instead of marshes transitioning to floodplain forests along tidal gradients⁸, the lower Amazon River Basin contains areas supplanting marsh for mangrove.

Mangroves are often associated with odd and intriguing community mixes — they grow atop coral reefs in Micronesia⁹, provide substrate for coral in the Caribbean¹⁰, and occur in isolated inland pockets as legacies of geologic time¹¹. On the other hand, not much is known about different tidal freshwater forested wetland associations outside of the United States, but if you look hard enough, they can be found¹².

